

User Manual

Step-by-Step Instructions for Playing the Multiplication Game

Step 1: Open the MARS Simulator.

Step 2: Load the Game Files

- Navigate to File > Open...
- Select all of the following .asm files and open them:
 - Main.asm
 - DrawBoard.asm
 - UpdateBoard.asm
 - MoveValidation.asm
 - CheckWin.asm
 - ComputerStrategy.asm
 - InitBoard.asm
 - Utils.asm
 - SysCalls.asm
- Make sure Main.asm is selected and active.

Step 3: Assemble the Game

- Go to Settings > Assemble All Files (or press F3).

Step 4: Run the Game

- Click the Go button or press F5 to run the game.

Step 5: Read the Welcome Message

- The console will display:
Welcome to the Multiplication Game!

Step 6: View the Initial Board

- A 6x6 grid of multiplication products will appear.
- Each number represents a unique product.

Step 7: Your Turn

You will see the message:
Your turn!

- Choose a number (1-9):
- Enter a number from 1 to 9 and press Enter.

Step 8: Move Validation

- If your input is invalid, you will see:
Invalid move! Try again.
- Enter a new number until it is accepted.

Step 9: Computer's Turn

The game will print:

Computer's turn!

- Computer selected: [number]
- The board updates after each turn.

Step 10: Visual Indicators

- Claimed cells show:
 - (P) for your claimed cell
 - (C) for the computer's claimed cell

Step 11: Sound Feedback

- A beep sound is played after each move.
- Two beeps are played when someone wins.

Step 12: Winning the Game

- When someone wins, you'll see:
You win!
or
Computer wins!

Step 13: Ending Manually

- The program will end automatically.
- You can stop the game early by pressing the red stop button in MARS.

Troubleshooting Tips

- If nothing displays, ensure all files are opened and assembled.
- If you receive errors, double-check the input range (1–9).
- Use the MARS console to follow prompts and feedback.

Enjoy the game and good luck winning against the computer!